Replication of *The Efficacy of Seven Gratitude Interventions for Promoting Subjective Well-Being* by Dang et al. (2025, *Journal of Positive Psychology*)

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80% Power Needs 93.79 (94) participants per group for a two-sided test at alpha = .05. This is 188 participants total. **5**

At 90% power, you need about 125 participants per group (total = 250) **5**

At 90% power, you need about 155 participants per group (total = 310) **5**
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Introduction

Dang et al. (2026) sought to test 7 gratitude interventions that are used in positive psychology to improve subjective well-being. These interventions included intrapersonal, interpersonal, and gratitude-to-God interventions. A key strength of this experiment is that most studies in the past used different measures of subjective well-being and only compared a few of the interventions against each other which made it impossible to make comparisons across interventions. By contrast, by including a comprehensive battery of interventions and standardized outcome measures (i.e., state gratitude, state positive/negative affect, and life satisfaction), this experiment allows for a direct comparison between gratitude interventions and can inform gratitude practices for the general public to incorporate into their lives.

In Study 1, it was found that writing a private gratitude letter was the most effective intervention in increasing positive affect. In this replication report, I want to replicate this finding and hope to find a similarly large Hedges' g of around .40 when comparing this intervention to an active control in a student population. This will hopefully expand the generalizability of the study and offer a new technique for students to try to increase their well-being.

Methods

Power Analysis

The key effect size for this replication (gratitude letter vs. events list) is Hedges' $g = .41$. Many power analysis programs only offer Cohen's d power estimates. With larger samples, the difference between Hedges' g and Cohen's d becomes minimal as the $J(x)$ correction factor applied to Cohen's d approaches 1 with large n. The conversion is as follows below:

Using the Hedges' g to Cohen's d conversion, we can divide Hedges' g by the $J(x)$ correction factor to recover Cohen's d, where $x = n_1 + n_2 - 2$ for the correction factor J. In Study 1, both conditions had exactly $n = 126$ participants, therefore $x = 126 + 126 - 2 = 250$.

$$
\begin{aligned}
J(x) \cdot d &= g \\
d &= \frac{g}{J(x)} \\
d &= \frac{g}{\frac{\Gamma(x/2)}{\sqrt{x/2} \times \Gamma((x-1)/2)}}
\end{aligned}
$$

$$
\begin{aligned}
&\#\text{Substitute } J(x) = J(250) \\
d &= g \times \frac{\sqrt{250/2} \times \Gamma((250-1)/2)}{\Gamma(250/2)} \\
d &= g \times \frac{\sqrt{125} \times \Gamma(124.5)}{\Gamma(125)} \\
d &= g \times 1.003013...
\end{aligned}
$$

```
# Calculate Correction Factor
J_250 <- sqrt(125) * gamma(124.5) / gamma(125)
J_250
```

```
[1] 1.003013
```

```
ComputedCohens_d = 0.41 * J_250
ComputedCohens_d
```

```
[1] 0.4112351
```

Now that we have the appropriate effect size of interest (0.4112351), we can use the `pwr` package to do a power analysis and calculate the required sample size for 80%, 90%, and 95% power.

```
# Step 1: Set CRAN as the source to install `pwr`
options(repos = c(CRAN = "https://cloud.r-project.org"))

# Step 2: Install `pwr`
install.packages("pwr")
```

```
The downloaded binary packages are in
    /var/folders/0p/p3phj575333gmy1dhsm2fsqr0000gp/T//RtmpSd6ztB/downloaded_packages
```

```r
# Step 3: Load `pwr` into the current session
library(pwr)
```

80% Power

```r
Power80 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .80,
           type = "two.sample",
           alternative = "two.sided")
```

```r
Power80$n
```

```
[1] 93.79321
```

At an alpha level of .05, for 80% power, we are going to need 93.7932103 participants per group. In total, that will be 188 participants.

90% Power

```r
Power90 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .90,
           type = "two.sample",
           alternative = "two.sided")
```

```r
Power90$n
```

```
[1] 125.2323
```

At an alpha level of .05, for 90% power, we are going to need 125.2322846 participants per group. In total, that will be 251 participants.

95% Power

```
Power95 = pwr.t.test(d = ComputedCohens_d,
           sig.level = .05,
           power = .95,
           type = "two.sample",
           alternative = "two.sided")
```

At an alpha level of .05, for 95% power, we are going to need 154.6466419 participants per group. In total, that will be 310 participants.

80% Power Needs 93.79 (94) participants per group for a two-sided test at alpha = .05. This is 188 participants total.

pwr.t.test(d = GToCohensD, sig.level = 0.05, power = 0.80, type = "two.sample", alternative = "two.sided")

At 90% power, you need about 125 participants per group (total = 250)

pwr.t.test(d = GToCohensD, sig.level = 0.05, power =.90, type = "two.sample", alternative = "two.sided")

At 90% power, you need about 155 participants per group (total = 310)

pwr.t.test(d = GToCohensD, sig.level = 0.05, power =.95, type = "two.sample", alternative = "two.sided")

This Section Was From Earlier Checkpoint. Can Skip Over This.

Relevance to Research Interests. In my personal research, I am interested in testing low-cost, accessible, and portable single-session interventions that can improve subjective well-being as well as uncover the mechanisms which underlie these changes to subjective well-being, particularly in student populations. Students represent a unique population

with higher rates of depression and anxiety as well as potentially limited financial resources to seek help. My research seeks to thus find preventative practices that can be used to buffer against the stressors of big life changes, coursework, and future career uncertainty. This experiment by Dang et al. (2025) has found that privately writing a gratitude letter that does not get shared with anyone has a sizeable impact on subjective well-being across a variety of outcomes including increasing gratitude and positive affect, as well as reducing negative affect. Interventions such as these are precisely the kinds of interventions I hope to design, expand upon, and test due to its brevity and financial accessibility.

Description of Stimuli and Procedures/Challenges. The original Dang et al. (2025) experiment tested 7 different interventions; however, I am most interested in the effect that the gratitude letter (vs. writing an events list active control) has on state positive affect. There was a larger effect of the gratitude intervention when compared to only measuring state positive affect in the control (Hedges' $g = .61$); however, as Haoyu suggested, having an active control that would be able to control for the effects that writing may have on positive affect would strengthen our ability to conclude that it was the gratitude aspect of writing that letter than lead to increases in happiness. This effect (gratitude letter vs. writing a list of events) was smaller but still quite large (Hedges' $g = .41$).

In the interest of replication, I will use the same instructions that were provided in the original study (Cunha et al., 2019; Dang et al., 2025; Emmons & McCullough, 2003; Walsh et al., 2022). Participants will be randomly assigned to two conditions and see the following directions:

- **Control Group: Events List (Cunha et al. 2019):** "In life, there are events, both large and small, that end up affecting us. Think back over the past day and write down five events that somehow affected you."
- **Experimental Condition: Gratitude Letter (Walsh et al., 2022):** "In our daily lives, we often thank people for both large and small things. However, it can sometimes be difficult to make time to express our gratitude more deeply. Today, we want you to write a letter of gratitude to someone who has done something for which you are extremely grateful. Please do not share your letter with this person or anyone else."

Participants will also be told:

> As you write, don't worry about perfect grammar and spelling, and remember that anything you write will remain strictly confidential. Should an experimenter read this entry in the future, it will be identifiable only by a participant number and not by your name'. (Dang et al., 2025).

Some potential challenges of replicating this study will be ensuring that participants actually complete the task, especially since data collection will occur online without supervision. To mitigate this, we will use `jsPsych` (de Leeuw et al., 2023) and prevent participants from using copy and paste functions. Of interesting note, the original Dang et al. (2025) study

did not exclude any participants from their analyses which reflects an intent-to-treat analysis approach (Ahn & Kang, 2023) which analyzes the data under unideal circumstances including non-compliance to reflect real-world treatment effects. In the interest of strict replication, I will follow the same procedure. I also think this is more interesting because often times, these positive psychology interventions are aimed at a large scale population (e.g., public school administrators recommending these strategies to students). Thus, non-compliance due to the nature of the task actually is inseparable from the outcome associated with the intervention. Simply analyzing the treatment effect under ideal conditions will hide these potential outcomes.

The key test in this study is a independent samples t-test. The Hedges' g will be a follow-up effect size calculation.

Reference List

Ahn, E., & Kang, H. (2023). Intention-to-treat versus as-treated versus per-protocol approaches to analysis. *Korean Journal of Anesthesiology*, *76*(6), 531–539. https://doi.org/10.4097/kja.23278

Cunha, L. F., Pellanda, L. C., & Reppold, C. T. (2019). Positive psychology and gratitude interventions: A randomized clinical trial. *Frontiers in Psychology*, *10*. https://doi.org/10.3389/fpsyg.2019.00584

Dang, A. V., Coles, N. A., Oishi, S., & McCullough, M. E. f(2026). The efficacy of seven gratitude interventions for promoting subjective well-being. *The Journal of Positive Psychology*, *21*(3), 517–535. https://doi.org/10.1080/17439760.2025.2502483

de Leeuw, J. R., Gilbert, R. A., & Luchterhandt, B. (2023). jsPsych: Enabling an open-source collaborative ecosystem of behavioral experiments. *Journal of Open Source Software*, *8*(85), Article 5351, https://joss.theoj.org/papers/10.21105/joss.05351.

Emmons, R. A., & McCullough, M. E. (2003). Counting blessings versus burdens: An experimental investigation of gratitude and subjective well-being in daily life. *Journal of Personality & Social Psychology*, *84*(2), 377–389. https://doi.org/10.1037/0022-3514.84.2.377

Walsh, L. C., Regan, A., Twenge, J. M., & Lyubomirsky, S. (2022). What is the optimal way to give thanks? comparing the effects of gratitude expressed privately, one-to-one via text, or publicly on social media. *Affective Science*, *4*(1), 82–91. https://doi.org/10.1007/s42761-022-00150-5

Learning Goals

Through this project, I hope to learn how to use jsPsych and SONA for data collection as well as get more familiarity with using Quarto to write academic reports. I also want to learn how to use R more (which I have begun to get better at since I have used it already in the power section). I think that this is a feasible project assuming that SONA has a sufficient amount of students because I have done similar projects in the past in undergrad (with less participants and no coding of the stimuli or data analysis) and those were completed quite reasonably fast.

I also hope this project will give me another experience with pre-registering hypotheses and making my analysis scripts/data open (I will completely redact people's lists and letters that they write, though).

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Methods

Power Analysis

Planned Sample

Aligned with the original study, we will analyze as an intent to treat and therefore not exclude participants. Aiming for 80% power, we will aim for 188 total participants with complete data. This should balance power and feasibility of collecting data.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.f